

Comparative Evaluation of Lexical Encoding Strategies in a Hybrid Chemical-Lexical Framework for Protein Function Prediction

Aaron Schröder

TH Bingen

Mat: 2194756

`aaron.schroeder@th-bingen.de`

August 30, 2025

Abstract

This study evaluates two computational approaches for predicting protein function by combining chemical similarity with lexical analysis of protein annotations. In the first approach, lexical vectors were generated using term frequency-inverse document frequency (TF-IDF) weighting, while in the second a binary encoding scheme was applied to mitigate concerns about frequency-driven biases. Protein neighbors were identified in a chemical structure space using an Euclidean k-d tree, and their functional annotations were compared through cosine similarity to retain only functionally consistent predictions.

Across 30 query proteins, both approaches achieved only partial success. For the TF-IDF model, 13.3% of predictions (4/30) were fully correct, 20.0% (6/30) were noisy but contained relevant terms, and 66.7% (20/30) were incorrect. The binary model showed a similar outcome, with 13.3% fully correct (4/30), 26.7% noisy (8/30), and 60.0% wrong (18/30).

Overall, both methods recovered biologically meaningful information in roughly one third of cases, but failed in the majority. These results highlight the potential of hybrid chemical-lexical approaches for function prediction while underscoring major challenges, particularly incomplete annotation coverage and the difficulty of reducing lexical noise.

1 Introduction

The rapid growth of public protein sequence databases has far outpaced experimental characterization, leaving most proteins functionally unannotated. Closing this annotation gap is a central challenge in computational biology, with implications for disease research, metabolic engineering, and drug discovery. Traditional approaches, such as annotation transfer from close homologs using BLAST or HMMER, often fail when no well-characterized relatives exist, highlighting the need for scalable methods that can infer function beyond strict sequence similarity.

In this work, we present a pipeline that integrates two complementary perspectives. A *chemical space*, derived from sequence similarity and physicochemical properties (SVD embeddings and Kidera factors), captures structural features, while a *lexical space*, built from Gene Ontology, InterPro, and curated protein descriptions, captures functional semantics. Query proteins are embedded into this hybrid space, and their function is inferred by aggregating annotations from their chemically and lexically consistent neighbors.

A critical component of this framework is the representation of annotations. While TF-IDF weighting emphasizes rare and specific terms, binary encoding provides a simpler presence/absence model less prone to spurious frequency effects. This study systematically compares the two strategies, addressing two key questions: (i) how the choice of encoding affects prediction accuracy, and (ii) what characteristic strengths and failure modes arise from each approach.

2 Related Work

Protein Similarity, Clustering, and Feature Extraction in UniRef50

The large scale of protein sequence space is evident in public datasets such as UniRef50, which clusters the UniProt database at 50% sequence identity and currently comprises tens of millions of unique clusters and hundreds of millions of proteins [6, 12]. Many clusters remain functionally uncharacterized, motivating the development of automated, scalable function prediction workflows [5, 6].

Homology-based tools, such as DIAMOND and BLASTp, enable rapid all-vs-all similarity calculations, crucial for mapping relations and inferring function in large datasets [4]. However, annotation transfer via sequence similarity faces classic hurdles [11]. Feature-based and hybrid workflows now incorporate dimensionality reduction and richer descriptors to supplement and sometimes supersede classical scoring matrices.

Kidera factors, a compact set of ten orthogonal descriptors derived by factor

analysis on 188 amino acid physicochemical properties, are now widely adopted as highly-informative features in such pipelines [9]. Combining similarity matrices with Kidera factors yields a "chemical space" that captures both global and local sequence properties.

Dimensionality Reduction and Construction of Chemical Spaces

Dimensionality reduction methods such as Singular Value Decomposition (SVD) [13] have become standard for extracting the principal axes of variation from large similarity matrices, resulting in tractable protein embeddings [4, 11]. In our approach, a high-dimensional similarity matrix (derived from DIAMOND/BLASTp) is projected into a space capturing the majority ($\sim 80\%$) of the sequence variance and augmented by the ten Kidera factors per protein. This composite feature vector enables efficient nearest neighbor queries and robust comparison in downstream tasks.

Lexical Analysis of Protein Annotations

Automated function prediction is further strengthened by analysis of protein annotation corpora, including Gene Ontology (GO) [1] terms, InterPro descriptions [8], and human-readable records from Swiss-Prot/TrEMBL [2]. Treating these annotations as text, they can be tokenized and analyzed as documents, allowing co-occurrence analysis and embedding via SVD [7].

This approach transforms sets of tokens (terms) into vector spaces in which semantically similar annotations lie close together. Document-level (protein-level) lexical vectors are then created as the aggregate of their component token vectors, a tactic shown to outperform simple binary or frequency encodings in many bioinformatics applications [7].

Hybrid Prediction in a Chemical–Lexical Space

Multi-modal workflows that integrate chemical (sequence-derived) and lexical (annotation-derived) feature spaces represent the current state of the art in protein function prediction [3, 11]. Given a query protein, the pipeline described here locates chemical neighbors in the reference database using Euclidean k-d tree searches on the SVD+Kidera chemical space embedding [10]. From these neighbors, annotation transfer is refined using cosine similarity in the lexical space, ensuring that only functionally and semantically coherent annotations are adopted.

Functional Gaps and Transfer Uncertainty

Despite recent progress, large swathes of UniProt (including UniRef50) remain "functionally dark"—clusters with little or no experimentally-verified annotation [5, 6]. This underscores the importance of empirical studies that systematically benchmark hybrid methods across annotation schemes, feature representations, and transfer strategies [11]. Evaluating the efficacy of vectorization strategies, within combined chemical/lexical spaces, remains an area with substantial room for empirical and methodological advancement.

3 Methodology

3.1 Data Preparation and Preprocessing

3.1.1 Reference Protein Dataset Construction

The reference dataset was constructed from the UniRef50 database [12], comprising approximately 50 million protein sequences. To ensure biological relevance and functional coverage, we specifically selected cluster reference proteins from a cluster with size greater than one. This reduced the amount of reference proteins to 23,483,433. All protein identifiers underwent canonicalization to maintain consistency across heterogeneous data sources using the standardization procedure described in Algorithm 1.

Algorithm 1 Protein Identifier Canonicalization

```
1: function CANONICALIZEID(identifier)
2:   Remove database-specific prefixes and version information
3:   Extract primary accession component
4:   Normalize capitalization and special characters
5:   return standardized_identifier
6: end function
```

3.1.2 Functional Annotation Processing

Three complementary annotation sources were processed to create structured corpora for subsequent lexical analysis (see algorithm 2):

Gene Ontology Annotations: We processed approximately 50 million protein-GO term associations from the Gene Ontology database. Terms were mapped to their human-readable descriptions, and the resulting corpus contained over 10,000 unique functional concepts after text normalization and filtering.

Protein Domain Annotations: Domain architecture information was extracted from approximately 45 million protein-domain associations. Raw domain descriptions were processed through semantic normalization, resulting in a curated set of over 34,000 unique domain descriptors.

Textual Protein Descriptions: Human-readable functional descriptions were processed from nearly 20 million protein-text pairs. After comprehensive text cleaning and tokenization, this corpus contained over 350,000 unique description elements suitable for semantic analysis.

Algorithm 2 Annotation Processing Pipeline

```
1: procedure PROCESSANNOTATIONS(annotation_data)
2:   Apply text normalization: remove special characters, normalize whitespace
3:   Perform corpus-specific tokenization
4:   Filter tokens by length and linguistic content
5:   return structured annotation corpus
6: end procedure
```

3.1.3 Query Protein Selection Strategy

Query proteins were selected using a stratified sampling approach designed to maximize functional diversity and coverage. The selection criteria prioritized:

- Representative coverage across functional categories
- Minimal sequence similarity within functional groups
- Comprehensive physicochemical property representation

This strategy yielded a final query set of 30 proteins representing the diversity of functional space while maintaining analytical tractability. These proteins are from *P.coccineus* and were analyzed using MapMans Mercator4 v7.0¹. The selected Proteins all belong to different level 1 bins according to MapMan ensuring maximum differences.

3.1.4 Quality Control Framework

A multi-tiered quality control framework was implemented throughout data preparation:

1. **Identifier Consistency:** Cross-dataset identifier mapping verification

¹https://www.plabipd.de/mercator_main.html

2. Coverage Assurance: Monitoring of annotation completeness across datasets

This rigorous preprocessing pipeline ensured both the quality of individual annotations and the consistency of their integration across multiple data sources, providing a robust foundation for subsequent analytical steps as shown in figure 1.

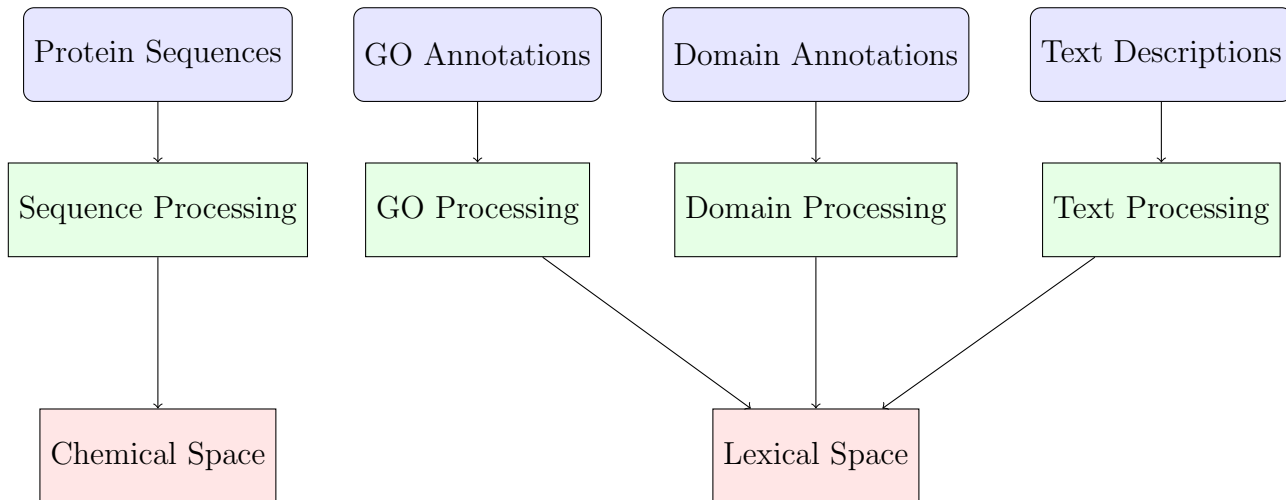


Figure 1: Conceptual framework of data integration showing the parallel processing of sequence information (chemical space) and functional annotations (lexical space).

3.2 Chemical Structure Space Construction

3.2.1 SVD-Based Protein Embedding

This work was contributed by Dennis Mendoza, TH Bingen

Protein sequence similarity was quantified using a geometric mean approach that incorporates both forward and reverse alignment scores to ensure symmetric similarity measures. For each protein pair (i, j) , the similarity score S_{ij} was computed as the geometric mean of four alignment metrics:

$$S_{ij} = \sqrt[4]{\text{overlap}_i \times \text{overlap}_j \times \text{pident}_i \times \text{pident}_j}$$

where *overlap* represents the alignment coverage and *pident* represents the percentage identity from DIAMOND/BLAST alignments.

The resulting sparse similarity matrix $\mathbf{S} \in \mathbb{R}^{n \times n}$ was centered using an implicit method to avoid constructing a dense, double-centered matrix. This was achieved by defining a linear operator that performs the centering operation during the matrix-vector products of the SVD solver:

$$\text{matvec_centered_op}(x) = \mathbf{S}x - \boldsymbol{\mu}x$$

where $\boldsymbol{\mu} \in \mathbb{R}^{1 \times n}$ is the row mean vector of $\mathbf{S}$, calculated as $\boldsymbol{\mu} = \frac{1}{n}\mathbf{S}\mathbf{1}$.

The truncated SVD was then computed on this implicit, centered matrix:

$$\mathbf{S}_{\text{centered}} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T$$

Dimensionality reduction was performed by retaining the top k components that collectively explain 80% of the total variance:

$$k = \min \left\{ m : \frac{\sum_{i=1}^m \sigma_i^2}{\sum_{i=1}^n \sigma_i^2} \geq 0.80 \right\}$$

where σ_i represents the i -th singular value. The resulting embedding $\mathbf{E} \in \mathbb{R}^{n \times k}$ captured the dominant patterns of structural similarity in a reduced-dimensional space.

3.2.2 Kidera Factor Computation

Physicochemical properties were quantified using Kidera factors [9], which represent ten orthogonal dimensions of amino acid properties derived from multivariate analysis of 188 physicochemical parameters. For each protein sequence, the Kidera factors were computed as:

$$\mathbf{k} = \frac{1}{L} \sum_{i=1}^L \mathbf{f}(a_i) \quad (1)$$

where L is the sequence length, a_i is the i -th amino acid in the sequence, and $\mathbf{k}(a_i)$ represents the precomputed Kidera factor vector for amino acid a_i using *kideraFactors()* from R’s *peptides* package.

Non-standard amino acids and ambiguous residues were handled through exclusion, with missing values imputed using the mean Kidera factor values across the reference dataset. This approach ensured robust representation of physicochemical properties while maintaining consistency across the protein set.

3.2.3 Combined Chemical Space

The structural embedding and physicochemical properties were integrated into a unified chemical space through concatenation as shown in figure 2:

$$\mathbf{C} = [\mathbf{E} \quad \mathbf{K}] \quad (2)$$

where $\mathbf{E} \in \mathbb{R}^{n \times k}$ represents the SVD-based structural embedding and $\mathbf{K} \in \mathbb{R}^{n \times 10}$ contains the Kidera factors for each protein.

The resulting chemical space captured both global structural similarities through the SVD embedding and local physicochemical properties through the Kidera factors, providing a comprehensive representation of protein characteristics relevant for functional prediction.

3.3 Lexical Analysis of Functional Annotations

3.3.1 Annotation Tokenization and Text Processing

Functional annotations from three distinct sources underwent specialized text processing pipelines tailored to their respective structural characteristics (fig. 3):

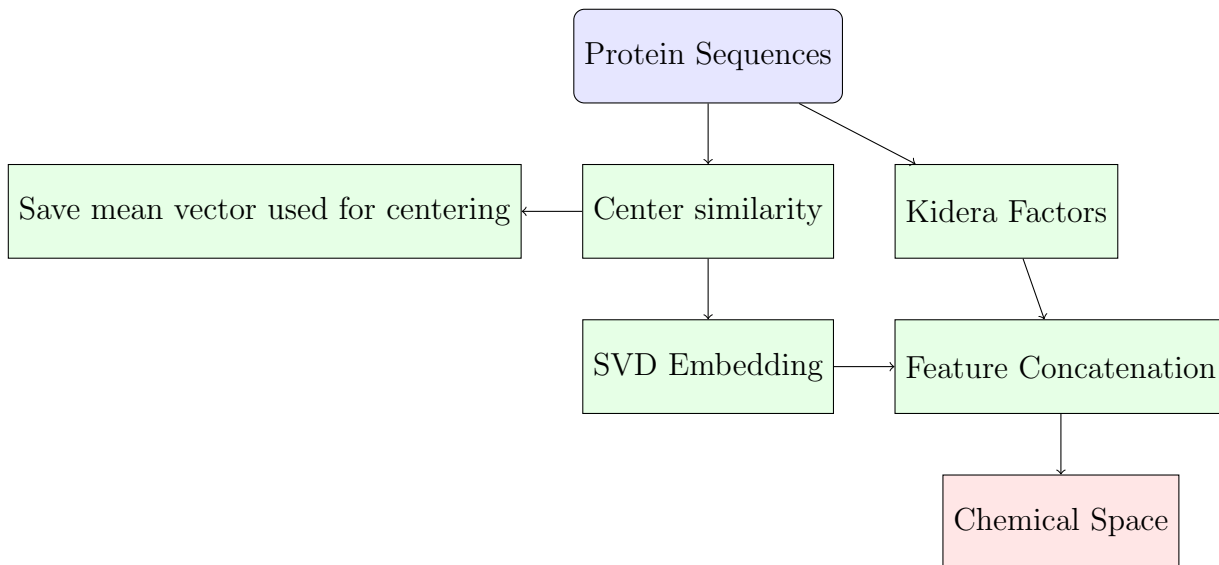


Figure 2: Chemical space construction pipeline showing integration of structural embeddings and physicochemical properties into a unified representation space.

Gene Ontology Annotations utilized pattern-based tokenization optimized for biological terminology. The tokenization process employed regular expressions designed to preserve complex biological concepts while separating compound terms:

$$\text{tokens} = \text{split}(\text{annotation}, [\backslash\text{s}, :]+) \quad (3)$$

InterPro Domain Annotations and **Human-Readable Descriptions** employed semantic splitting on punctuation marks and natural language boundaries:

$$\text{tokens} = \text{split}(\text{annotation}, [\backslash\text{s}, ;,]+) \quad (4)$$

All tokenization processes (algorithm 3) incorporated comprehensive text normalization including:

- Removal of special characters and formatting artifacts
- Case normalization to lowercase
- Elimination of stop words; An additional check is performed for descriptions exceeding three words. The final token is validated against a dictionary; if it's not a recognized word (e.g., "Protein", "Subunit"), it's likely a non-standard identifier (e.g., "IL_231") and is removed to prevent issues with TF-IDF weighting.
- Length-based filtering (retaining tokens > 2 characters)
- Linguistic filtering (requiring alphabetic content)

This processing resulted in curated vocabularies of 10,503 unique GO terms, 34,584 unique domain descriptors, and 353,389 unique textual description elements respectively.

Algorithm 3 Annotation Tokenization Pipeline

```
1: procedure TOKENIZEANNOTATION(annotation, corpus_type)
2:   if corpus_type = GO then
3:     pattern  $\leftarrow$  [\s.,;]+
4:   else
5:     pattern  $\leftarrow$  [\s.,;]+
6:   end if
7:   tokens  $\leftarrow$  SPLIT(annotation, pattern)
8:   tokens  $\leftarrow$  FILTER(tokens, length > 2)
9:   tokens  $\leftarrow$  FILTER(tokens, contains_letters)
10:  tokens  $\leftarrow$  NORMALIZE(tokens, lowercase)
11:  return tokens
12: end procedure
```

3.3.2 TF-IDF Encoding Approach

The primary method implemented a Term Frequency-Inverse Document Frequency (TF-IDF) encoding. However, the SVD was performed on the term-frequency (TF) co-occurrence matrix for computational efficiency. The IDF weights were applied in the subsequent document projection step.

The term-frequency weighted matrix $\mathbf{X}_{\text{tf}} \in \mathbb{R}^{n \times m}$ was used to build the co-occurrence matrix:

$$\mathbf{C} = \mathbf{X}_{\text{tf}}^T \mathbf{X}_{\text{tf}}$$

Singular Value Decomposition was applied to $\mathbf{C}$ to obtain the semantic word axes:

$$\mathbf{C} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

Dimensionality reduction retained components explaining 90% of the total variance, with a minimum of $k = 4$ dimensions for Gene Ontology and $k = 2$ for other corpora:

$$k = \min \left\{ m : \frac{\sum_{i=1}^m \sigma_i^2}{\sum_{i=1}^n \sigma_i^2} \geq 0.90 \right\}, \quad k_{\text{GO}} \geq 4, \quad k_{\text{IPR,HRD}} \geq 2$$

The resulting projection matrix $\mathbf{U}_k \in \mathbb{R}^{m \times k}$ was used to embed documents into the reduced semantic space by projecting the TF-IDF weighted document-term matrix:

$$\mathbf{D}_{\text{tf-idf}} = \mathbf{X}_{\text{tf-idf}} \mathbf{U}_k$$

where $\mathbf{X}_{\text{tf-idf}}$ is the sparse matrix with entries $\text{tf}(t, d) \times \text{idf}(t)$.

3.3.3 Binary Encoding Approach

The secondary encoding method employed a binary representation of annotation presence/absence, constructing a document-term matrix $\mathbf{X} \in \{0, 1\}^{n \times m}$ where:

$$x_{ij} = \begin{cases} 1 & \text{if protein } i \text{ contains annotation } j \\ 0 & \text{otherwise} \end{cases}$$

The co-occurrence matrix $\mathbf{C}$ was computed as:

$$\mathbf{C} = \mathbf{X}^T \mathbf{X}$$

Singular Value Decomposition was applied to $\mathbf{C}$ to obtain the semantic space:

$$\mathbf{C} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

Dimensionality reduction retained components explaining 85% of total variance, with different corpus-specific minimum dimensions as the TF-IDF approach:

$$k = \min \left\{ m : \frac{\sum_{i=1}^m \sigma_i^2}{\sum_{i=1}^n \sigma_i^2} \geq 0.85 \right\}, \quad k_{\text{GO}} \geq 5, \quad k_{\text{IPR,HRD}} \geq 3$$

The resulting projection matrix $\mathbf{U}_k \in \mathbb{R}^{m \times k}$ was used to embed proteins into the reduced-dimensional lexical space:

$$\mathbf{D} = \mathbf{X} \mathbf{U}_k$$

where $\mathbf{D} \in \mathbb{R}^{n \times k}$ represents the lexical embedding of proteins.

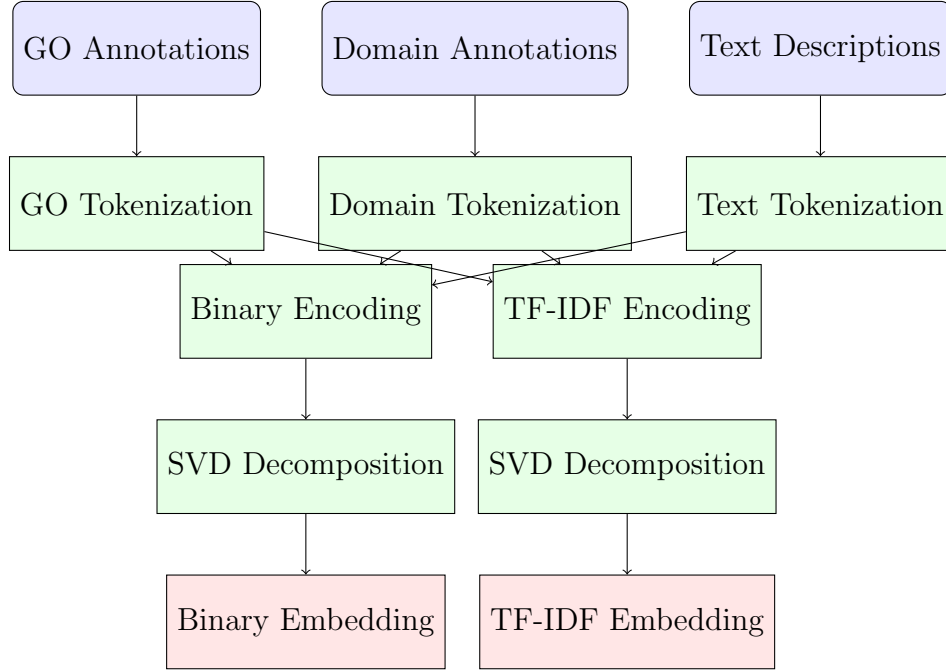


Figure 3: Lexical analysis pipeline showing parallel processing of annotation sources through both binary and TF-IDF encoding strategies, culminating in semantic embeddings for each approach.

The binary encoding approach prioritized annotation presence over frequency, potentially offering advantages for biological applications where the existence of functional features may be more informative than their frequency of occurrence. The TF-IDF baseline provided a standard information retrieval comparison to evaluate this hypothesis.

3.4 Query Protein Embedding and Function Prediction

3.4.1 BLAST-Based Weighting and Normalization Scheme

Query protein embedding utilized sequence similarity information from DIAMOND/BLAST alignments against the reference database. A critical aspect of the pipeline was ensuring that the query embedding process used the same normalization as was applied during the construction of the reference chemical space via SVD.

For each query protein q and reference protein r , a geometric mean similarity score $G(q, r)$ was first computed from four alignment metrics:

$$G(q, r) = \sqrt[4]{\text{overlap}_f \times \text{overlap}_r \times \text{pident}_f \times \text{pident}_r}$$

This geometric mean incorporates both forward (f) and reverse (r) alignment directions, considering both alignment coverage (overlap) and sequence identity (pident).

To ensure consistency with the centered SVD of the reference space, the weighting scheme employed the precomputed mean similarity vector $\boldsymbol{\mu} \in \mathbb{R}^n$ (where μ_r is the mean similarity of reference protein r across all proteins in the reference set). The final, normalized weight $w(q, r)$ was calculated as:

$$w(q, r) = \frac{G(q, r) - \mu_r}{\sum_{r' \in R} G(q, r')}$$

This normalization strategy subtracts the reference-specific mean μ_r from the raw similarity score $G(q, r)$ before sum normalization. This process precisely mirrors the implicit centering operation ($\mathbf{S} - \boldsymbol{\mu}$) performed during the SVD of the reference similarity matrix, ensuring that the query protein is embedded into the chemical space using the same coordinate system as the reference proteins.

3.4.2 Chemical Space Projection

Query proteins were embedded into the chemical space through weighted combination of their similar reference proteins:

$$\mathbf{v}_q^{\text{chem}} = \sum_{r \in N(q)} w(q, r) \cdot \mathbf{v}_r^{\text{chem}} \quad (5)$$

where $N(q)$ represents the set of reference proteins with significant BLAST hits to query q , $\mathbf{v}_r^{\text{chem}}$ denotes the chemical space vector of reference protein r , and $w(q, r)$ represents the normalized alignment weight.

The query’s intrinsic Kidera factors $\mathbf{k}_q$ were incorporated through concatenation:

$$\mathbf{v}_q^{\text{combined}} = [\mathbf{v}_q^{\text{chem}} \quad \mathbf{k}_q] \quad (6)$$

This combined representation captured both structural similarities inferred from sequence alignment and intrinsic physicochemical properties.

3.4.3 Nearest Neighbor Retrieval

The k -nearest neighbors in chemical space were identified using Euclidean distance metric and efficient KD-tree search:

$$N_k(q) = \arg \min_{r \in R} \left\{ \left\| \mathbf{v}_q^{\text{combined}} - \mathbf{v}_r^{\text{combined}} \right\|_2 \right\} \quad (7)$$

with $k = 50$ neighbors retained for subsequent lexical consistency filtering. This approach ensured computational efficiency while maintaining comprehensive coverage of the local chemical neighborhood.

3.4.4 Lexical Consistency Enforcement

A sequential consistency check was applied to ensure functional coherence among retrieved neighbors. For each corpus $c \in \{\text{GO}, \text{IPR}, \text{HRD}\}$, the ordered list of neighbors from the KD-tree search was processed. Proteins were added to the consistent set S_c only if their lexical vector was sufficiently similar to every protein already in the current set S_c .

This greedy algorithm initializes S_c with the first (chemically closest) neighbor. For each subsequent candidate neighbor r_i in the ordered list, its lexical vector $\mathbf{v}_{r_i}$ is compared to the vector of every protein already in S_c using cosine similarity. The candidate is only admitted if all pairwise similarities meet or exceed the threshold θ .

The mean document vector $\mathbf{d}_c$ is calculated as the average of all vectors in S_c but is not used for the admission decision; it is computed and stored for downstream functional annotation aggregation after the consistent set is finalized.

Algorithm 4 Lexical Consistency Filtering

Require: Ordered neighbor list $N = [r_1, r_2, \dots, r_{50}]$, lexical vectors $\mathbf{V}_c$, default threshold $\theta = 0.9$

Ensure: Consistent protein set S_c , mean document vector $\mathbf{d}_c$

```

1:  $S_c \leftarrow \{r_1\}$                                 ▷ Initialize with first neighbor
2:  $\mathbf{D} \leftarrow \{\mathbf{v}_{r_1}\}$                         ▷ Initialize document vector set
3: for  $i \leftarrow 2$  to 50 do
4:    $\mathbf{v} \leftarrow \mathbf{v}_{r_i}$                             ▷ Current candidate vector
5:    $\text{sim}_{all} \leftarrow \{\cos(\mathbf{v}, \mathbf{v}_s) \mid \forall s \in S_c\}$     ▷ Compare to all in current set
6:   if  $\min(\text{sim}_{all}) \geq \theta$  then
7:      $S_c \leftarrow S_c \cup \{r_i\}$ 
8:      $\mathbf{D} \leftarrow \mathbf{D} \cup \{\mathbf{v}\}$ 
9:   else
10:    break                                           ▷ Terminate on first inconsistency
11:  end if
12: end for
13:  $\mathbf{d}_c \leftarrow \text{mean}(\mathbf{D})$                         ▷ Compute mean for aggregation
```

This approach (figure:4) ensures that all proteins in the consistent set S_c are pairwise functionally coherent, maintaining a minimum cosine similarity $\theta = 0.8$ (default is 0.9, for research purposes 0.8 was used in the tests. Except for single cases all similarities were still greater than 0.9) between every member. The algorithm (4) terminates upon encountering the first candidate that fails this pairwise test, resulting in a functionally cohesive cluster.

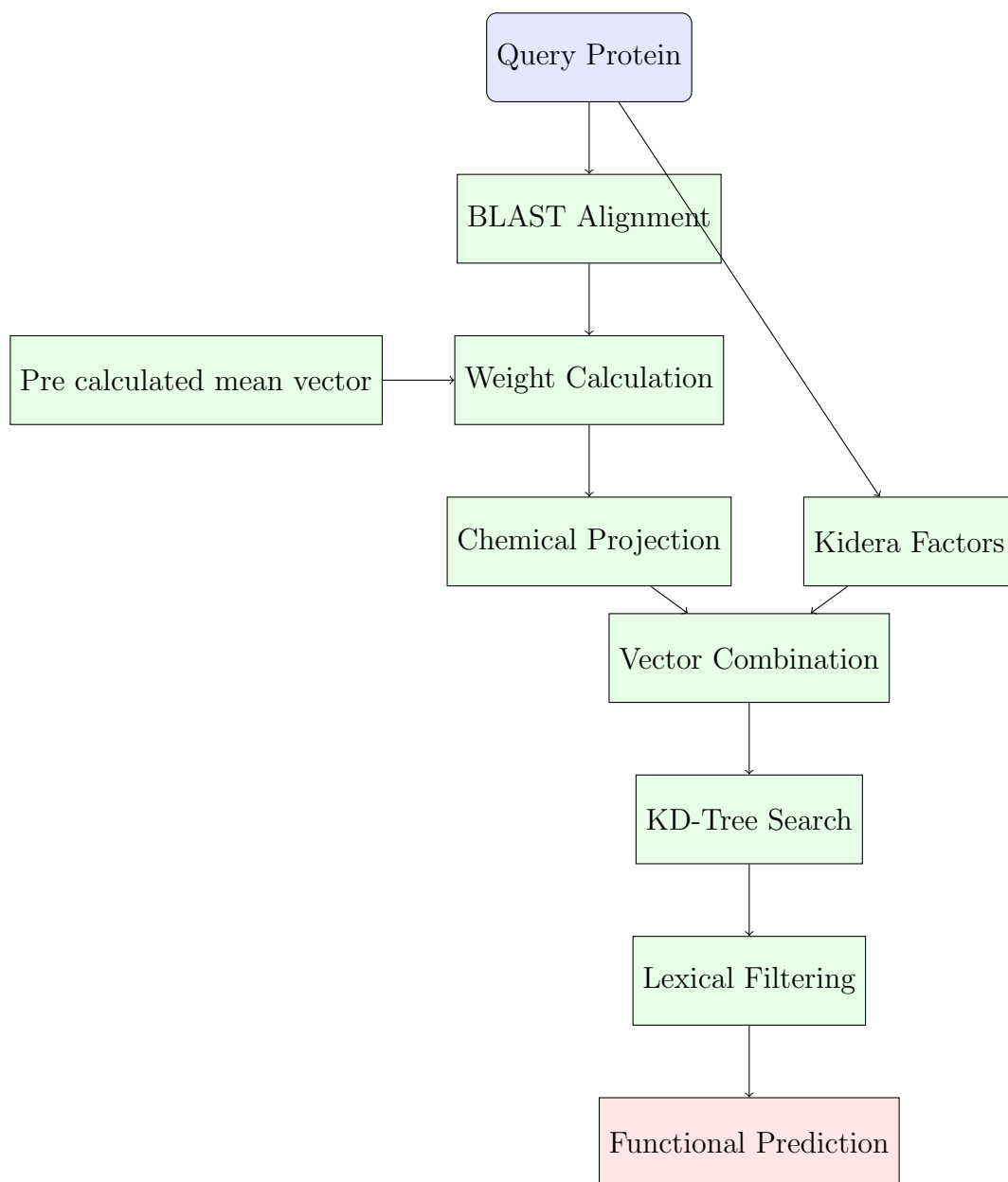


Figure 4: Query processing pipeline showing the integration of sequence alignment, chemical space projection, and lexical consistency filtering for functional prediction.

The final functional prediction for query protein q was derived from the consistent set S_c through aggregation of annotations weighted by their lexical similarity to the mean document vector $\mathbf{d}_c$, providing a robust and coherent functional profile based on both structural and functional evidence.

3.5 Methodological Rationale

The design choices for this protein function prediction pipeline were driven by the need to bridge the gap between sequence-based similarity and functional annotation, while accounting for the inherent complexities and inconsistencies of biological data. Below we outline the reasoning behind our key methodological decisions.

3.5.1 Choice of Euclidean Distance in K-D Tree Construction

We utilized a Euclidean K-D tree for the initial nearest-neighbor search in the high-dimensional chemical feature space, as opposed to a spherical (angular) metric. The Euclidean distance was chosen because it directly measures the absolute magnitude of difference between the feature vectors derived from protein sequences. This is appropriate as we intend to capture proteins with the smallest overall quantitative deviation in their chemical descriptor space. A spherical metric, which measures the angle between vectors and is the basis for cosine similarity, is more suitable for measuring orientation rather than magnitude. Since the initial retrieval step is intended to find the most chemically proximate proteins in absolute terms, Euclidean distance provides a more intuitive and physically meaningful metric for this stage of the pipeline.

3.5.2 Decoupling of Chemical and Lexical Spaces

The pipeline explicitly separates the chemical (sequence-based) and lexical (text-based) similarity measures. This design reflects a fundamental hypothesis: the most sequence-similar protein is not always the most informative source for functional annotation. By first retrieving a set of candidates based purely on chemical similarity and then re-ranking them based on the semantic consistency of their annotations, the pipeline can potentially correct for cases where the top sequence hit has poor, missing, or misleading annotations. This two-stage process allows the model to leverage the strengths of both approaches: the broad recall of sequence similarity and the precise, functional specificity of text-based analysis.

3.5.3 Use of Cosine Similarity for Textual Annotation

Cosine similarity was selected as the metric for comparing annotation vectors due to its effectiveness in text mining and its inherent property of measuring orientation rather than magnitude. This makes it robust to differences in the number of annotations per protein. A protein with many GO terms is not automatically deemed more similar to the query than a sparsely annotated one; instead, the measure focuses on the conceptual overlap between the two sets of terms. This is particularly advantageous in biological contexts where annotation depth is highly variable across proteins.

3.5.4 Evaluation of Binary and TF-IDF Encoding

We implemented and compared both binary and TF-IDF (Term Frequency-Inverse Document Frequency) encoding schemes to evaluate their respective strengths and weaknesses for this specific task.

- Binary Encoding provides a baseline by treating all annotated terms as equally important. It measures the pure overlap of functional descriptors.
- TF-IDF Encoding adds a layer of weighting that reflects the specificity of a term. Terms that are common across many proteins are down-weighted, while rare, more specific terms are given higher importance.

This comparison allows us to test the hypothesis that weighting terms by their specificity will lead to more precise and biologically relevant predictions by reducing noise from common, generic annotations.

3.5.5 Selection of Level 1 MapMan Bins as Test Data

The test set was constructed using proteins from diverse Level 1 MapMan bins (e.g., photosynthesis, stress response, protein synthesis) rather than a random selection. This strategic choice ensures that the benchmark evaluates the pipeline across a wide spectrum of distinct biological functions. A random set could be skewed towards overrepresented functional categories. By selecting representatives from high-level bins, we explicitly test the model’s ability to discriminate between fundamentally different biological processes, providing a more rigorous and generalizable assessment of its performance.

3.5.6 Utilization of GO, InterPro, and HRD Corpora

The three annotation corpora—Gene Ontology (GO), InterPro (IPR), and Homology-derived Reference Descriptions (HRD)—were selected for their complementary strengths:

- GO provides a standardized, hierarchical vocabulary of functional terms, allowing for a rich semantic description of a protein’s role.
- InterPro offers a consolidated view of protein domains and families, identifying functional modules and evolutionary relationships.
- HRD provides concise, expert-curated, full-protein functional descriptions from databases like Swiss-Prot, which are often considered the gold standard.

Using all three in parallel allows the pipeline to synthesize evidence from different types of functional evidence (ontology, domains, and curated summary), while also accounting for possibly missing annotations in single corporas.

3.5.7 Normalization of Similarities Prior to SVD

The similarity matrix of the blast all against all comparison was normalized before undergoing Singular Value Decomposition (SVD). This step is critical to ensure that the variance in the data reflects true biological patterns rather than artifacts of the data structure. Normalization adjusts for differences in scale between the similarity scores. Without it, a corpus with generally higher similarity scores could disproportionately influence the dimensionality reduction process, potentially overshadowing more subtle but biologically important patterns from other corpora. Normalization creates a level playing field, allowing SVD to identify the most significant latent factors based on the relative structure of the data, not its absolute magnitude.

3.6 Manual Evaluation and Comparative Analysis

3.6.1 Evaluation Framework Design

A comprehensive manual evaluation framework was established to assess the quality and biological relevance of the functional predictions. The evaluation encompassed three distinct comparative dimensions:

1. **Intra-corpus consistency:** Analysis of annotation coherence within each functional corpus (GO, InterPro, HRD)
2. **Inter-corpus comparison:** Assessment of complementarity and agreement between different annotation sources
3. **MapMan benchmark validation:** Direct comparison against curated functional classifications from the MapMan database

Functional predictions were considered correct if they matched the MapMan classification at appropriate hierarchical levels, with partial credit assigned for predictions at parent or child categories in the functional hierarchy.

This comprehensive evaluation methodology enabled rigorous assessment of prediction quality, identification of systematic strengths and weaknesses across different annotation types, and validation against established biological knowledge, providing insights into the practical utility of the developed functional prediction pipeline.

3.6.2 Code Availability

The entire protein function prediction pipeline, including data preprocessing, chemical space construction, lexical analysis, and neighbor retrieval, was implemented in Python and R.

The source code, detailed installation instructions, configuration files, and all parameters necessary to reproduce the results of this study are publicly available under an open-source license in the following GitLab repository:

URL: https://gitlab.rlp.net/a.hallab/tensor-omics/-/tree/Protein_Function_Prediction?ref_type=heads

4 Results

4.1 Performance Evaluation of Annotation Encoding Models

A comparative analysis was conducted between two text encoding methodologies—binary encoding and TF-IDF (Term Frequency-Inverse Document Frequency)—for protein function prediction. The evaluation assessed three key aspects: (1) intra-corpus consistency, referring to the agreement among predicted terms within each annotation corpus (GO, IPR, HRD); (2) inter-corpus consistency, measuring agreement between different corpora; and (3) functional accuracy, determining concordance with expert-curated MapMan annotations.

Predictions were classified into three distinct categories:

1. **Correct prediction:** Consistent, correct annotations across all corpora with minimal noise
2. **Noisy prediction:** Correct or biologically relevant annotations accompanied by substantial incorrect predictions

3. **Wrong prediction:** No functionally relevant annotations matching the MapMan designation

Model	Correct	Noisy	Wrong
Binary encoding	4 (13.3%)	8 (26.7%)	18 (60.0%)
TF-IDF encoding	4 (13.3%)	6 (20.0%)	20 (66.7%)

Table 1: Performance comparison of encoding methodologies across 30 proteins

4.2 Annotation Consistency Analysis

The evaluation revealed distinct patterns of annotation consistency across prediction corpora. HRD annotations demonstrated superior intra-corpus consistency, typically providing singular, specific functional descriptions derived from homology data. IPR annotations exhibited moderate to low internal consistency, frequently containing mixtures of relevant domains alongside phylogenetically unrelated domain predictions. GO annotations displayed variable consistency, ranging from coherent functional sets to biologically disjoint term collections.

Inter-corpus analysis showed only 33.3% of predicted annotation sets achieved high consistency across GO, IPR, and HRD predictions. The majority (66.7%) exhibited low inter-corpus consistency, with contradictory predictions across annotation sources. HRD annotations consistently provided the most reliable functional predictions, serving as the most accurate singular annotation source.

4.3 Case Studies of Characteristic Prediction Patterns

4.3.1 Optimal Performance Case (pc1675)

Both encoding methodologies performed optimally for photosynthesis-related protein pc1675. All predicted terms across GO, IPR, and HRD corpora consistently referenced chlorophyll binding, photosynthesis, and light harvesting processes, achieving near-perfect cosine similarity scores (around 0.999). The predictions matched both MapMan annotation ("component of LHC-II complex") and SwissProt reference ("Chlorophyll a-b binding protein 21, chloroplastic OS=*Nicotiana tabacum*"). All 32 predicted proteins across corpora showed perfect functional concordance, demonstrating the model's capability to identify orthologous functions across taxonomic boundaries (here: *P. cocineus* to *N. tabacum*).

4.3.2 TF-IDF Superior Performance Case (pc302)

For glutathione S-transferase cluster pc302, TF-IDF encoding demonstrated superior performance. While both models identified limited correct annotations (1/6 in IPR, 1/5 in HRD), TF-IDF successfully retrieved the specific SwissProt reference matching MapMan's annotation. Binary encoding failed to identify this critical reference. The TF-IDF model achieved this despite substantial noise in GO predictions, which erroneously suggested ribosomal functions, highlighting its improved capacity for relevant term identification amidst noise.

4.3.3 Binary Encoding Superior Performance Case (pc196)

For calcium sensor protein pc196, binary encoding outperformed TF-IDF methodology. Binary encoding identified one calcium-related GO term among two predictions, while TF-IDF failed to

identify any calcium-related terms in its singular GO prediction. Both models showed poor performance in IPR and HRD predictions, with TF-IDF generating particularly discordant annotations (including viral polymerase and integrase domains) despite high cosine similarity scores (0.86-0.93), indicating potential over-weighting of rare but irrelevant terms.

4.3.4 Systematic Failure Case (pc483)

Both models failed completely for NAC transcription factor pc483, consistently predicting phospholipase D functionality throughout all corpora. TF-IDF encoding amplified this error by generating additional noise (8 vs. 2 IPR predictions), including completely unrelated domains such as integrases and endonucleases. This case demonstrates a fundamental limitation in sequence-based annotation transfer when functional domains overlap between unrelated protein families.

4.3.5 Differential Noise Profile Case (pc1020)

For pepsin-type protease pc1020, both models identified correct aspartic peptidase annotations but with substantial noise. Binary encoding produced membrane-related noise (4/7 predictions), while TF-IDF generated secretion system-related noise (3/6 predictions) due to enhanced weighting of bacterial secretion terms. This case illustrates how encoding methodologies differentially affect error profiles, with TF-IDF amplifying rare but specific incorrect terms versus binary encoding’s tendency toward generalistic incorrect annotations.

4.3.6 Annotation Coverage as a Source of Prediction Error

A critical vulnerability identified within the prediction pipeline stems from inconsistent annotation coverage across different biological databases. The algorithm’s initial step retrieves the top 50 nearest protein neighbors based on chemical structure similarity via a K-D tree search. Subsequently, functional predictions are generated by performing cosine similarity calculations on the textual annotations of these proteins. However, the pipeline’s performance is critically dependent on a fragile assumption: that the sequence-similarity ranking correlates with both the availability and functional relevance of annotations.

This assumption fails when annotation coverage is incomplete or uneven across corpora. The case of the calcium sensor protein pc196 provides a definitive example of this failure mode:

- The K-D tree correctly identified the 50 most chemically similar proteins.
- However, for the Gene Ontology (GO) corpus, the first and second most sequence-similar proteins (ranks #1 and #2) lacked GO annotations entirely.
- The algorithm was therefore forced to select the third-ranked protein as the initial reference for the cosine similarity calculation. This protein’s GO annotation (“RNA-directed RNA polymerase activity”) was functionally orthogonal to the true calcium sensing function.
- The next available GO annotation appeared at sequence-rank #5 (“calcium ion binding”), but the calculation was already biased by the incorrect reference from rank #3.

These differences become clearer when focusing on Figure 5. In the GO corpus (Fig. 5a), 30% of cases (9/30) show that the top hit identified by the K-D tree in chemical space lacks any available annotation and therefore cannot be used. In three of these instances, only the third K-D tree hit could be utilized. Annotation coverage is somewhat higher in the IPR corpus, where 77% of the best K-D tree hits are annotated, and in the remaining 23% the second hit had to be used (Fig. 5b).

The highest coverage is observed in the HRD corpus (Fig. 5c), with only a single top hit lacking annotation. These findings are consistent with the overall annotation densities of the databases: Gene Ontology covers 12,236,003 of 23,483,433 entries (52%), IPR covers 14,044,515 entries (59%), and HRD provides annotations for 20,046,013 entries (85%).

This scenario creates a fundamental inconsistency: different corpora may base their predictions on reference proteins with vastly different degrees of chemical similarity to the query. For instance, the HRD corpus might utilize the protein at sequence-rank #1, while the GO corpus is forced to use a protein at sequence-rank #3. This corpus-specific starting point inherently decouples the functional predictions from the underlying chemical similarity structure, leading to contradictory and erroneous predictions across annotation sources.

The gold standard case (pc1675) succeeded precisely because annotation coverage was complete—the top sequence-similar proteins were fully annotated across all corpora, allowing the chemical and functional similarity spaces to align perfectly. The high error rates observed across the benchmark are therefore not merely a function of incorrect annotations, but more fundamentally a consequence of incomplete and uneven annotation coverage in biological databases, which the current pipeline architecture is not designed to mitigate. This represents a critical limitation for any method that relies on database annotations for protein function prediction.

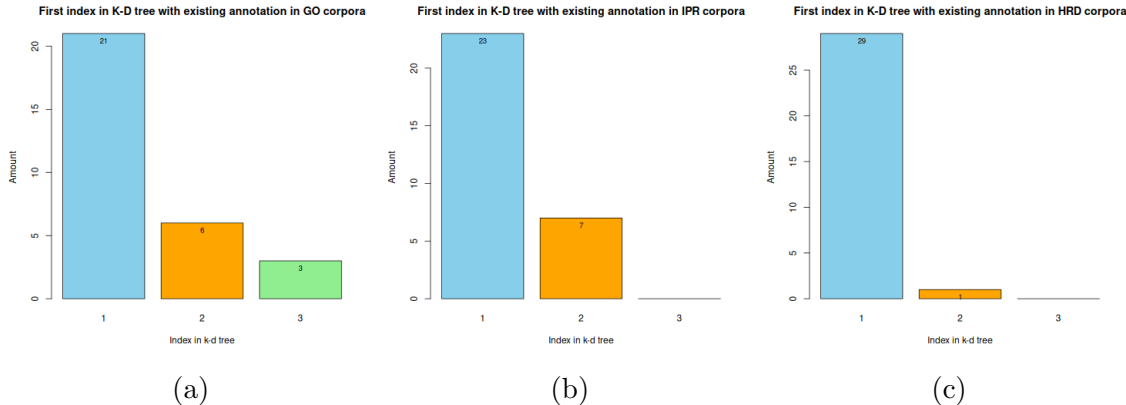


Figure 5: K-D tree index of the first protein with available annotation

5 Discussion

The present study demonstrates that a hybrid pipeline combining chemical embeddings with lexical annotation analysis can recover meaningful protein function predictions even in the absence of close sequence homologs. The integration of structural information, encoded through SVD-derived embeddings and Kidera factors, with annotation-derived semantic vectors provides a versatile framework that balances recall and specificity. Nevertheless, the results also reveal limitations and point to several opportunities for methodological refinement and deeper biological interpretation.

5.1 Engineering and Methodological Perspective

From an engineering perspective, the modular design of the pipeline is a clear strength, as it explicitly separates chemical similarity from lexical similarity and combines them only at the stage of annotation transfer. This ensures that sequence-derived information does not automatically dominate the predictions and allows poor or incomplete annotations to be filtered by semantic coherence. At the same time, several aspects of the implementation could be improved.

One area for refinement concerns the treatment of physicochemical descriptors. The concatenation of SVD embeddings with Kidera factors may introduce scale biases related to sequence length or amino acid composition. Centering and normalizing the Kidera factors prior to integration would mitigate such effects and lead to more balanced representations.

The comparative analysis of annotation encodings further highlights methodological challenges. While TF-IDF captured highly specific and discriminative terms, it also amplified rare spurious annotations, whereas binary encoding produced more robust but less precise results. This suggests that a hybrid encoding strategy could be superior, for example by applying TF-IDF weighting only to sufficiently frequent terms while treating rare terms in a binary manner. A related limitation lies in the inclusion of uninformative tokens such as “protein”, “domain”, or “uncharacterized protein”, which add semantic noise without contributing functional insight. Explicit blacklisting or global down-weighting of such terms would likely sharpen the lexical space. Finally, the current tokenization process relies on simple, largely uniform patterns across the different corpora. This approach can be further optimized to preserve more contextual information and generate biologically meaningful tokens, thereby improving the semantic quality of the co-occurrence matrix and the downstream lexical representations.

Another methodological consideration concerns the lexical similarity threshold. At present, a strict cosine cutoff enforces functional coherence but may be overly rigid in borderline cases. Allowing for dynamic relaxation, combined with ontology-aware generalization, would make the predictions more robust: when confidence is low, it may be more appropriate to report a broader Gene Ontology category (e.g. “metabolic process”) instead of a fine-grained but uncertain term (e.g. “glycolytic process”).

In future work, these improvements should be coupled with more rigorous benchmarking. While the present study relied on manual evaluation against MapMan classifications, quantitative metrics such as precision, recall, F_1 score, and semantic similarity measures in the GO hierarchy would allow more systematic comparisons across different encoding schemes and prediction strategies.

5.2 Biological Perspective

From a biological standpoint, the evaluation across thirty diverse proteins provides insight into when and why the pipeline succeeds or fails. Overall, both encoding strategies failed to recover biologically meaningful predictions in roughly two thirds of the cases. For the binary encoding, 13.3% of predictions were fully correct, 26.7% were noisy but still contained biologically relevant terms, and 60.0% were incorrect. The TF-IDF approach achieved the same proportion of fully correct predictions (13.3%) but produced slightly fewer noisy predictions (20.0%) and a higher fraction of wrong predictions (66.7%). These results indicate that both encoding methods capture functional signals, yet they differ systematically in their balance between precision and stability.

The source of these differences lies primarily in how the respective models construct document vectors. TF-IDF has the advantage of emphasizing rare, discriminative terms while down-weighting highly frequent and generic ones. This property allows it to correctly identify functionally specific annotations that would otherwise be obscured by common terms. However, this same weighting

scheme can also create artificial structure in the semantic space by overemphasizing rare terms that are not biologically informative, thereby introducing misleading noise. The binary model, in contrast, reduces this risk by treating all terms equally, which helps preserve clusters of related annotations. Yet, because it lacks a mechanism to distinguish informative from generic descriptors, it is more strongly influenced by common tokens such as “protein”, “domain”, or “subunit”. As a result, the false positives generated by the binary model tend to reflect broad, generic signals, whereas the false positives in the TF-IDF model more often arise from spurious but specific terms.

These complementary error profiles highlight an inherent trade-off in lexical encoding for biological annotation: TF-IDF increases the chance of retrieving rare, functionally precise terms but at the cost of sensitivity to misleading noise, while binary encoding yields more stable predictions dominated by common signals but sacrifices functional resolution.

When high-quality reference annotations were available, the methods reproduced functions with remarkable specificity. The photosystem protein (pc1675) exemplifies this, where both encoding schemes correctly identified chlorophyll-binding and photosynthesis-related roles with near-perfect cosine similarity, fully aligning with MapMan and SwissProt annotations. This demonstrates that the hybrid chemical-lexical framework is capable of bridging species boundaries and transferring function across evolutionary distances.

However, the case studies also exposed systematic weaknesses. The calcium sensor protein (pc196) highlighted the tendency of TF-IDF to overweight rare but misleading terms, while binary encoding produced broader, less specific but sometimes more stable annotations. Such complementary error profiles illustrate how the choice of encoding scheme directly influences biological interpretability. More generally, annotation coverage emerged as the dominant bottleneck. Because only about half of the reference proteins were annotated with GO terms (52%), 59% with InterPro domains, and 85% with HRD descriptions, the most chemically similar neighbors often lacked usable functional annotations. In these situations, the pipeline was forced to rely on more distant proteins, which introduced contradictory or misleading terms.

Even when correct general functions were identified, fine-grained specificity was frequently lost. For instance, proteases were consistently predicted as hydrolases, but additional unrelated annotations, such as secretion system terms, inflated the results. This reflects a trade-off between broad process-level accuracy and precise domain-level resolution. The variability across annotation corpora further emphasizes this limitation. HRD consistently provided the most coherent and reliable predictions, while InterPro annotations often contained mixtures of relevant and irrelevant domains, and GO annotations showed highly variable internal consistency. This indicates that biological success depends not only on the algorithm but also on the choice and completeness of annotation sources.

5.3 Outlook

Taken together, these findings suggest that further progress will require simultaneous advances on both the methodological and biological fronts. On the engineering side, normalization of descriptors, hybrid encoding strategies, ontology-aware generalization, and adaptive neighbor retrieval offer concrete paths for improvement. On the biological side, more consistent annotation coverage and the systematic handling of incomplete or uninformative entries will be critical. The excellent performance observed in cases with complete annotations highlights the promise of the approach. The challenge now lies in extending this reliability to the much larger set of proteins that remain sparsely or inconsistently annotated, thereby contributing to the reduction of the annotation gap in the protein universe.

References

- [1] Michael Ashburner, Catherine A Ball, Judith A Blake, David Botstein, Heather Butler, J Michael Cherry, Allan P Davis, Kara Dolinski, Selina S Dwight, Janan T Eppig, et al. Gene ontology: tool for the unification of biology. *Nature genetics*, 25(1):25–29, 2000.
- [2] Amos Bairoch and Rolf Apweiler. The swiss-prot protein sequence data bank and its supplement trembl. *Nucleic acids research*, 25(1):31–36, 1997.
- [3] Frimpong Boadu, Ahhyun Lee, and Jianlin Cheng. Deep learning methods for protein function prediction. *Proteomics*, 25(1-2):2300471, 2025.
- [4] Benjamin Buchfink, Klaus Reuter, and Hajk-Georg Drost. Sensitive protein alignments at tree-of-life scale using diamond. *Nature methods*, 18(4):366–368, 2021.
- [5] Janani Durairaj, Andrew M Waterhouse, Toomas Mets, Tetiana Brodiazhenko, Minhal Abdullah, Gabriel Studer, Mehmet Akdel, Antonina Andreeva, Alex Bateman, Tanel Tenson, et al. What is hidden in the darkness? deep-learning assisted large-scale protein family curation uncovers novel protein families and folds. *bioRxiv*, pages 2023–03, 2023.
- [6] Janani Durairaj, Andrew M Waterhouse, Toomas Mets, Tetiana Brodiazhenko, Minhal Abdullah, Gabriel Studer, Gerardo Tauriello, Mehmet Akdel, Antonina Andreeva, Alex Bateman, et al. Uncovering new families and folds in the natural protein universe. *Nature*, 622(7983):646–653, 2023.
- [7] Rafael S Gonçalves, Jason Payne, Amelia Tan, Carmen Benitez, Jamie Haddock, and Robert Gentleman. The text2term tool to map free-text descriptions of biomedical terms to ontologies. *Database*, 2024:baae119, 2024.
- [8] Sarah Hunter, Rolf Apweiler, Teresa K Attwood, Amos Bairoch, Alex Bateman, David Binns, Peer Bork, Ujjwal Das, Louise Daugherty, Lauranne Duquenne, et al. Interpro: the integrative protein signature database. *Nucleic acids research*, 37(suppl_1):D211–D215, 2009.
- [9] Akinori Kidera, Yasuo Konishi, Masahito Oka, Tatsuo Ooi, and Harold A Scheraga. Statistical analysis of the physical properties of the 20 naturally occurring amino acids. *Journal of Protein Chemistry*, 4(1):23–55, 1985.
- [10] Rina Panigrahy. Nearest neighbor search using kd-trees. *citeseerx.ist.psu.edu*, 2006.
- [11] Riccardo Percudani and Carlo De Rito. Predicting protein function in the ai and big data era. *Biochemistry*, 64(11):2345–2352, 2025.
- [12] Baris E Suzek, Yuqi Wang, Hongzhan Huang, Peter B McGarvey, Cathy H Wu, and UniProt Consortium. Uniref clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics*, 31(6):926–932, 2015.
- [13] Michael E Wall, Andreas Rechtsteiner, and Luis M Rocha. Singular value decomposition and principal component analysis. In *A practical approach to microarray data analysis*, pages 91–109. Springer, 2003.

6 Appendices

AI was used to improve the text in clarity and scientific tone.

A full list of results is provided in GitLab: https://gitlab.rlp.net/a.hallab/tensor-omics/-/tree/Protein_Function_Prediction?ref_type=heads. The full list of materials is only available on the BioServer due to the size of the files.

Versions of the scripts that were used to generate the output evaluated in this paper:

- calculate_kidera.R: Version 1.2.7
calculates the kidera factors of the reference protein set
- calculate_query_kidera.R: Version 1.0.0
Calculates the kidera factors of a given file; used for the query proteins
- create_annotation_table_optimized.R: Version 1.3.2
Creates the annotation tables (direct mappings protein_id $\rightarrow$ annotation)
- create_chemical_space.R: Version 1.3.2
Uses the kidera factors and the svd results to build the chemical space
- create_lexical_axes.R: Version 1.4.3
Creates the lexical axes with tf-idf weighting
- create_lexical_axes_no_idf.R: Version 1.4.4
Creates the lexical axes with binary weighting
- create_named_mean_vector.py: Version 1.0.0
Merges the saved mean vector to proteins IDs to create a named vector
- extract_annotations.R: Version 2.1.3
Merges Protein.IDs to annotations based on coprora in the order of the K-D tree results
- extract_annotations_no_idf.R: Version: 1.1.3
Merges Protein.IDs to annotations based on coprora in the order of the K-D tree results
- run_query.R: Version 4.3.3
Runs the query with a given set of parameters and input files

The used material was downloaded on:

- Swissprot: 18.06.2025
- Trembl: 18.06.2025
- UniRef50: 23.04.2025
- Gene Ontology: 11.06.2025